EuroCTP Consolidated Tape Multicast feed external specifications

Version 1.2

March 2026

1. Content and document versioning

1.1. Content

This document contains EuroCTP's real-time consolidated tape technical specifications for its FIX SBE feed delivered via direct access to EuroCTP's data centres.

1.2. Document versioning

Date	Version	Changes
20/12/2025	v 1.0	Initial document publication
23/01/2026	v 1.1	Typos corrected in page 41 MDQualityIndicator (3105) values are 0, 1, 2 or 4 and page 42 TrdRegTimestamp_2 (769) field name corrected. MMT flag construct description amended on page 13, inclusion in first release to be confirmed.
12/03/2026	V1.2	Post-trade message Added that the MDOriginType (1024) is omitted when LastMkt (30) is set to 'XOFF' or 'SINT' Confirmation that the uncrossing bid and offer prices will be in scope for go live.
23/03/2026	V1.3	Order matching system status Message name aligned to TradingSessionStatus [h] in the second table of paragraph 4.2.13

2. Definitions and acronyms

2.1. Acronyms

APA Approved Publication Arrangements
TV Trading Venues (includes Regulated Markets and Multilateral Trading Facilities)
RM Regulated Market
MTF Multilateral Trading Facility
MIC Market Identifier Code
ETFs Exchange Traded Funds
CLOB Central Limit Order Book
ESMA European Securities and Markets Authority
FBA Frequent Batch Auction

2.2. Definitions

Continuous order books	are trading systems that by means of an order book and a trading algorithm operated without human intervention matches sell orders with buy orders on the basis of the best available price on a continuous basis, pursuant to RTS1 Annex I definition of Continuous order book trading system.
Composites	Shares and ETFs that are aggregated into a set of Composites based on the following two characteristics, the ISIN code of the share or ETF and its Trading Currency.

Data contributor	is a Trading Venue or APA that is identified by its operating MIC within ESMA's List of data contributors to the equity CTP
Exchange Traded Funds	Article 4(1)(43) of Directive 2014/65/EU 'exchange-traded fund' means a fund of which at least one unit or share class is traded throughout the day on at least one trading venue and with at least one market maker which takes action to ensure that the price of its units or shares on the trading venue does not vary significantly from its net asset value and, where applicable, from its indicative net asset value;
Frequent Batch Auctions systems	are order books which match orders exclusively on the basis of a periodic auction trading system within the definition of Table 1 of Annex I of Delegated Regulation (EU) 2017/587 (RTS 1)
Instruments	Shares and ETFs that are aggregated into a set of Instruments based on the following three characteristics, the ISIN code of the share or ETF, its Trading Currency and its place of Listing
MMT	MMT is an industry led market standard Market Model Typology for transactions.
Periodic auction trading systems	are trading systems that match orders on the basis of a periodic auction and a trading algorithm operate without human intervention, pursuant to RTS1 Annex I definition of Periodic auction trading system.
Place of Listing	A trading venue where an issuer of a financial instrument has requested or approved the trading or admission to trading of its financial instrument on a d trading venue per the definition of field 8 'Request for admission to trading by issuer' in Table 3 of Delegated Regulation (EU) 2017/585 (RTS 23)
RTS on input and output data of consolidated tapes	Chapters I and II of supplementing Regulation (EU) No 600/2014 of the European Parliament and of the Council with regard to regulatory technical standards specifying the input and output data of consolidated tapes, the synchronisation of business clocks and the revenue redistribution by the consolidated tape provider for shares and ETFs, and repealing Delegated Regulation (EU) 2017/574.

3. EuroCTP real-time market data feed access

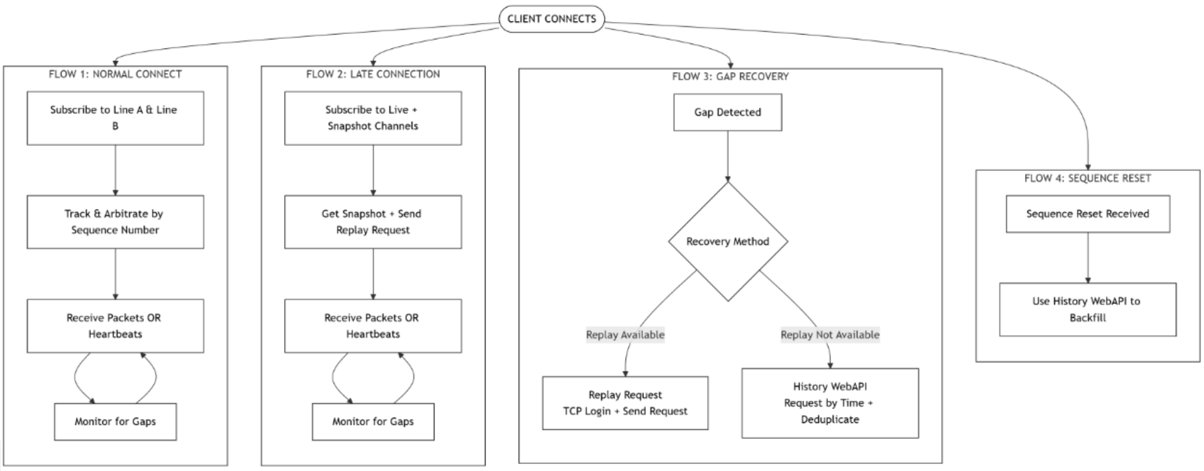
EuroCTP offers its clients access to its real-time feed via four different delivery methods The specifications are the same for the directly connected clients and those sourcing the data from the web API.
This document only covers the Direct connection to EuroCTP FIX SBE feeds, noting that EuroCTP uses the FIX Latest standards.

Modes of Delivery			
			xx Publication date DD/MM/YYYY
Access	Description	Delivery mechanism	Syntax

Direct connection to EuroCTP Ticker Plant *(or via a market data vendor)*	Binary feed	Multicast	FIX SBE
Internet	Binary feed ISO 20022 compliant feed	Web API EuroCTP website	FIX SBE
	File downloads	Web API On demand download via the EuroCTP website	CSV
	Display	On demand via the EuroCTP website	N/A

3.1. EuroCTP Ticker plant direct access

EuroCTP Ticker Plant delivers real-time market data over a number of load-balanced IP multicast channels.
Description of the client communication flow



<u>Subscribing to the data feed</u>

Consuming the data
Connection is established by subscribing to both EuroCTP's Lines A and B. Once established, the user may track and arbitrate outbound messages using their Sequence Numbers. Users may monitor gaps through received packets and heartbeat messages.

Late connection
In the case of late connection, and once EuroCTP Lines A and B have been subscribed, the user may consume the Snapshot message to establish the latest pre-trade data information and the Replay Request message in order to receive all post-trade information since the start of the day.

Gap recovery

In the case where a user wishes to receive missing data, it can send a replay request. This will enable full post-trade data retrieval and partial pre-date data retrieval.
Should the user require additional data, it can login to the TCP channel and retrieve the data.

- In the case where the login request is successful, replay requests can be sent. These include the sequence number range required by the user, packets will be sent with the requested sequence number range. The line will then be dropped after timeout.
- In the case where the login request was unsuccessful, EuroCTP will send a Logout message, indicating that invalid credentials were provided, the session is dropped.

Sequence reset
In the case where EuroCTP has to resent the message sequence numbers, EuroCTP will publish a Sequence Reset Message. Users wishing to retrieve missing data will be able to do so using EuroCTP's WebAPI.

3.1.1. Multicast channels

EuroCTP will use multicast to disseminate market data. The feed will be organised in several multicast channels. EuroCTP will use Feed ID and Channel ID to uniquely identify each channel. Packet header will contain information about Feed ID and Channel ID.
For Post-Trade data EuroCTP will only provide incremental updates. For Pre-trade and Regulatory Status feed EuroCTP will provide Incremental updates as well as snapshots feed. Snapshot channels have the same structure as Incremental updates channels.

Feed ID represents type of the feed. Possible values:

- 1 = Pre-trade feed, including European Best Bid Offer (EBBO), European Best Auction Price (EBAP) and European Frequent Batch Auctions (EFBA)
- 2 = Post-trade feed
- 3 = Regulatory Status feed

Channel ID identifies specific multicast channel / partition within each feed. Below table contains

Channel IDs and Feed IDs associated with each partition \ channel.

Asset class	Instrument/ Composite	Feed type	Feed ID	Channel ID
Shares	Instrument	Incremental Updates - Pre-trade	1	1
Shares	Instrument	Snapshots - Pre-trade	1	129
Shares	Instrument	Incremental Updates - Post-trade	2	51
Shares	Composite	Incremental Updates - Pre-trade	1	2
Shares	Composite	Snapshots - Pre-trade	1	130
Shares	Composite	Incremental Updates - Post-trade	2	52
ETFs	Instrument	Incremental Updates - Pre-trade - sub channel 1	1	3
ETFs	Instrument	Snapshots - Pre-trade - sub channel 1	1	131
ETFs	Instrument	Incremental Updates - Pre-trade - sub channel 2	1	4
ETFs	Instrument	Snapshots - Pre-trade - sub channel 2	1	132
ETFs	Instrument	Incremental Updates - Pre-trade - sub channel 3	1	5
ETFs	Instrument	Snapshots - Pre-trade - sub channel 3	1	133
ETFs	Instrument	Incremental Updates - Pre-trade - sub channel 4	1	6
ETFs	Instrument	Snapshots - Pre-trade - sub channel 4	1	134

ETFs	Instrument	Incremental Updates - Post-trade	2	53
ETFs	Composite	Incremental Updates - Pre-trade - sub channel 1	1	7
ETFs	Composite	Snapshots - Pre-trade - sub channel 1	1	135
ETFs	Composite	Incremental Updates - Pre-trade - sub channel 2	1	8
ETFs	Composite	Snapshots - Pre-trade - sub channel 2	1	136
ETFs	Composite	Incremental Updates - Pre-trade - sub channel 3	1	9
ETFs	Composite	Snapshots - Pre-trade - sub channel 3	1	137
ETFs	Composite	Incremental Updates - Pre-trade - sub channel 4	1	10
ETFs	Composite	Snapshots - Pre-trade - sub channel 4	1	138
ETFs	Composite	Incremental Updates - Post-trade	2	54
All	All	Incremental Updates - Regulatory Status	3	76
All	All	Snapshots - Regulatory Status	3	204
All	All	Recovery – Post-trade	2	250

3.1.2. Recovery and replay services

Clients needing to recover missing data may do so by submitting Replay Request message via EuroCTP's dedicated TCP channel, which has been set up specifically for this purpose.

The replay responses will be sent via that same TCP channel. The message formats are described in section 5.2 of this document.

Connection details for packet retransmission requests will be provided in the client onboarding pack.

Start of the day

EuroCTP clients are expected to connect to its consolidated tape feeds prior to the opening of the service, subscribing to both Line A and Line B live channels. Both Line A and Line B transmit identical message streams. Each packet includes a packet-level sequence number that allows the client to identify duplicates. EuroCTP clients must perform packet arbitration using this sequence number and process only the earliest received packet

If a client connects after the opening of the service, it should first subscribe to the live data feed to start processing live data, and then, in order to reintegrate missing data and backfill trade history, the following steps must be taken.

- Reintegrating pre-trade and regulatory data: listening to EuroCTP's Snapshot channel in order to process the next Snapshot message.
- Backfilling trade history from start of the day: by sending a Replay Request message by setting BeginSeq to 0 and setting EndSeq to the last received Packet Sequence number.

Client can use message level sequence numbers (instrument level sequence numbers) to determine the correct order between live updates and Snapshot messages for each instrument and take the latest (with the higher sequence number).

During the course of the day

EuroCTP provides two independent lines for each Channel, Line A and Line B. Each line contains the identical messages. Clients should subscribe to both lines, and reconcile messages using packet-level sequence number.
Clients will track the packet sequence numbers from both live channels in order to detect potential packet losses.

EuroCTP provides Version of Sequence Number in additional to the packet level sequence number, this parameter can be used to detect potential loss of Sequence Reset messages.

If a gap is detected on both Lines, clients may request the resubmission of lost packets via a Replay Request message.

- Pre-trade and regulatory data can be requested going back to the last 10 million sequence numbers.

- Post-trade data can be requested going back to the first transaction of that day.

Should a client request pre-trade and regulatory data going back further than the last 10 million sequence numbers, the request will still be accepted and the last 10 million messages will be sent back via the dedicated response channel.

Replay Request message rejection

BeginSeq and EndSeq are mandatory fields in the Replay Request message. If these fields are not specified EuroCTP will respond with a Logout message indicating that the session is terminated, the Logout message will contain a logout reason code indicating the root cause session termination.

EuroCTP will respond with a Logout message if values are negative.

EuroCTP will send a Resend Start message indicating actual BeginSeq and EndSeq that will be resent. EuroCTP will send a Logout message once the resend is complete. The Logout message will contain the logout reason code (successful completion of the resend request). BeginSeq and EndSeq can be different from what was initially requested, these will indicate what was actually resent.

EuroCTP will send Resend Start message with BeginSeq equals to EndSeq followed by a Logout message with logout reason (successful completion of the resend request), in case the provided values are outside the supported interval.

EuroCTP does not support more than one Replay Request message for each channel. If a second Replay Request for the same client and channel is received while the processing of the first Replay Request message has not yet been completed, EuroCTP will send a Logout message with an error code indicating that the request limits have been breached, and terminate the session.

Historical data

If a client is not able to retrieve the data through the Replay Request mechanism, it can use EuroCTP's WebAPI to request historical data based on the requested time interval in order to backfill their data.

Message identifiers such as Transaction Identification Code, EBBO identification number, EBAP identification number and EFBA identification number should be used to remove duplicates.

EuroCTP failover or system restart

In the cases where EuroCTP's systems were restarted, or failed over to another instance, Sequence Reset messages will be sent, indicating that sequence number for subsequent packages will start from zero.

EuroCTP does not support Replay Requests for pre-trade and regulatory data messages sent before the sequence reset event, in such cases clients should use EuroCTP's WebAPI historical data service to backfill their data.

In case of outages, EuroCTP will resend the post-trade data that may have been dropped during the failover process via a separate recovery channel once the Sequence Reset message has been sent, as a consequence the same transaction message may be sent several times, and Transaction Identification codes should be used to remove duplicates.

3.1.3. Client access

Access to multicast channels is controlled on network level. Additional information will be requested during client onboarding.
Replay functionality is available via a dedicated TCP channel. Clients will receive a login and password that they use to authenticate when establishing a TCP connection. Authentication is performed on Application level using FIX Logon message.

4. Binary feed format (FIX SBE)

4.1. Message overview

4.1.1. Message types

4.1.1.1. Administrative messages

Administrative messages are used to control the flow. All these messages have SchemaID=2

Administrative message	Template ID	Description
Heartbeat	1	This message will be sent whenever no data messages have been sent for 1 second
Sequence Reset	2	This message will be sent when EuroCTP initiates a packet level sequence reset, indicating that the packet level sequence number will be reset to 0
Logon	10	This message is used to authenticate while connecting to the EuroCTP replay channel
Logout	11	This message will be sent by the EuroCTP server when the connection has dropped
Replay Request	20	This message is used to request missed data
Replay Start	21	This message is used to indicate that the processing of the Replay Request has started
Snapshot End	30	This message will be sent once a snapshot cycle has been completed

4.1.1.2. Application messages

Application messages are used to transfer Market Data. All these messages have SchemaID=1

Application message	Template ID	Description
EBBO	101	European Best Bid and Offer messages contain the European best bid and offer price for Instruments and Composites made available to trade on continuous lit order books and periodic auction books.
EBAP	102	European Best Auction Price messages contain the European highest and lowest periodic auction price, volume weighted auction price and total volume and volume weighted auction price for Instruments and Composites made available to trade on continuous lit order books that operate periodic auctions.
EFBA	103	European Frequent Batch Auction Price messages contain the European highest and lowest periodic auction price, volume weighted auction price and total volume and volume weighted auction price for Instruments and Composites made available to trade on order books that operate frequent batch auctions.

Post-trade	201	Transactions executed on European Trading Venues or reported to European APAs.
Instrument status	301	Status of instruments made available to trade on European Trading Venues.
Order matching system status	302	Status of European Trading Venues order matching systems.

4.1.2. Data types

Data type	Length	Description
AppId	32 bytes	Alphanumerical 32 bytes
AppVersion	16 bytes	Alphanumerical 16 bytes
Currency	3 bytes	Alpha 3
Decimal	9 bytes	Used for Price and Size, represented as mantissa int64 and exponent int8, up to 17 decimal spaces
ESMA codes	4 bytes	Alpha 4
InternalIdentifier	8 bytes	int64
ISIN	12 bytes	Alphanumerical 12
LogoutReason	256 bytes	Alphanumerical 256 bytes
MIC	4 bytes	Alpha 4
MMT flag	14 bytes	Alphanumerical 14
Password	32 bytes	Alphanumerical 32 bytes
RptGrpHeader	2 bytes	Represents SBE repeating group header, consists of two fields - entryLength uint8 - the fixed-length size of one group entry's fixed fields, and numOfEntries uint8 - number of entries in the repeating group
SeqNum	8 bytes	int64
Specific flag (out of sequence etc…)	1 bit	Flags combined in a single field
Timestamp	8 bytes	Nanoseconds since EPOCH
TransactionIdentifier	52 bytes	Alphanumerical 52 bytes
Username	32 bytes	Alphanumerical 32 bytes

MMT efficient encoding

The MMT flag is included in EuroCTP's FIX SBE feed and is presented in its efficient encoding 14 character string. It, will be constructed by EuroCTP, based on the information in the Data Contributors' input messages, following the MMT 5.0 standard and will be available in subsequent releases.

Each character represents a position starting at position 1. When a mandatory position does not have a value, it is represented with a '?'

4.1.3. Timestamps

Timestamps are in nanoseconds since epoch 1st Jan 1970, (up until 1st Jan 2036 when Epoch will be rebased).

4.1.4. Numbers

EuroCTP will be sending all numbers in little-endian byte order - least significant part stored at the beginning.

4.1.5. Packet header

Packet headers will include a packet sequence number and dissemination time in the following formats.

Name	Type	Description
EncodingType	Uint16	Identifier of the encoding used in the message payload. Always little-endian will be used. Value is 0xEC79 (60537) in little-endian encoding.
PacketSize	Uint16	Overall packet size including packet header
FeedID	Uint8	Feed ID of the feed the message belongs to
ChanelID	Uint8	Channel ID of the feed channel the message belongs to
PacketSeqVersion	Uint8	Version of the Sequence Number
PacketSeqNum	Uint64	Packet Sequence Number Each Multicast channel will have its own set of monotonically increasing sequence numbers
Dissemination Time	Uint64	Nanoseconds from start of Epoch
Name	Type	Description

In the event of a system failover the sequence number will reset to zero, a dedicated message is sent prior to the reset, please refer to the Administrative messages section of this document.

4.1.6. Framing header

Messages will contain the following header following the SBE Standard:

Name	Type	Description
MessageLength	Uint16	Overall message length including headers to support framing.

4.1.7. Unit header

Messages will contain the following header following the SBE Standard:

Name	Type	Description
MsgSize	Uint16	The length of the entire message including the message header reported in number of bytes.
TemplateID	Uint16	Template ID, represents message type. Refer to the Message Type section to obtain concrete Template IDs used for each message type
SchemaID	Uint16	Identifies the schema to which the message belongs (Administrative Messages, Application Messages) 1 = Application Messages 2 = Administrative Messages
Version	Uint16	Schema version used for backward compatibility

4.1.8. Logout Reason Codes

Logout Code	Logout Reason	Description
0	Success	The session was successfully finished
1	Bad Encoding	Framing header encodingType field has invalid value
2	Unexpected Message	The message received is not expected now
3	Bad Message Length	Framing header messageLength field is out of range
101	Invalid Credentials	Logon failed due to invalid credentials provided
102	Too Many Connections	Client's limit on active connections is exhausted
103	InvalidChannelID	ChannelID provided withing ReplayRequest is invalid

104	TooManyReplayRequests	Client's limit on active replay request for the channel is reached
105	InvalidReplayRequest	Replay request is malformed
201	BadSequenceVersion	Provided Sequence Version does not match the current version of the sequence number
999	Other Error	Represents errors not covered by concrete error code already mentioned

4.2. Message structure

4.2.1. Heartbeat

If no new messages were sent in the channel for more than one second, EuroCTP will generate heartbeat messages. A heartbeat message will be generated and sent every second if no messages have been published during that 1 second interval. The heartbeat package will not increase the packet level sequence number for this channel. It will contain a sequence number that is equal to the heartbeat of the last non-heartbeat message.

Clients can detect disconnect or lost packages by observing sequence numbers on package level of events received in each channel. Heartbeat messages will allow client to detect disconnect or lost packages even during quiet periods. By tracking the last processed package sequence number and the sequence number in the current package, clients will be able to detect gaps and request replay data for the affected channel.

Heartbeat messages conform to the standard SBE packet and message header structure but contain no message payload. Each heartbeat is transmitted as a stand-alone message within its own packet. The packet-level sequence number is not incremented for heartbeat packets; instead, it retains the sequence number of the last packet that contained a business message.

4.2.2. Logon message

Logon messages are used to authenticate when connecting to the replay channel via TCP. Client are requested to provide Client App Id and Client App Version in Logon messages in order to be used to identify the source of connection. These can be used to communicate with EuroCTP's support team.

The message should include the following information

Name	Type	Description
Username	Username	Client Username used to authentication
Password	Password	Client Password used to authentication
ClientAppId	AppId	Client Application Id
ClientAppVersion	AppVersion	Client Application Version

4.2.3. Logout message

Logout messages are used to notify the client when a session terminates. The EuroCTP server will send logout messages before disconnecting the session after processing replay requests, or due to inactivity and in case of any errors, logout messages contain the reason for the logout.

Name	Type	Description
LogoutCode	Uint16	Logout Code representing root cause for session termination. Refer to Logout Reason Codes section for detail
LogoutReason	LogoutReason	Logout Reason explaining the reason for logout. Refer to Logout Reason Codes section for detail

4.2.4. Replay Request message

The message will need to include the following:

Name	Type	Description
Feed ID	Uint8	Feed ID for which the replay request is being sent
Channel ID	Uint8	Channel ID for which the replay request is being sent
PacketSeqVersion	Uint8	Version of Packet level sequence number. Must be the same as sequence number version received in sequence reset message
BeginSeqNo	SeqNum	Packet level sequence number of the first message in range to be resent
EndSeqNo	SeqNum	Packet level sequence number of the last message in range to be resent

4.2.5. Replay Start message

Indicates EuroCTP is starting to process a Replay Request. The message will contain the actual Start and End packet level sequence numbers that will be resent to the client.

Name	Type	Description
PacketSeqVersion	Uint8	Version of Packet level sequence number.
BeginSeqNo	SeqNum	Reject Code representing root cause for rejection. Refer to Error Codes section for detail

EndSeqNo	SeqNum	Reject Reason explaining the reason for rejection. Refer to Error Codes section for detail

4.2.6. Snapshot End message

In order to enable clients, that are establishing connection to the EuroCTP feed during the course of the day EuroCTP periodically sends latest snapshot of pre-trade data and regulatory status. Each channel containing incremental updates will have corresponding snapshot channel that will contain snapshots. Snapshot End message represents the end of snapshot transmission.
Snapshot End message conforms to the standard SBE message header structure but contain no message payload. Feed ID and Channel ID are provided in packet header.

4.2.7. Sequence reset message

Indicates that the packet level sequence number will be reset for the Channel. Packet level sequences are managed independently for each channel. The message contains the Sequence Number version that will be used.

Name	Type	Description
PacketSeqVersion	Uint8	Version of Packet level sequence number.

4.2.8. European Best Bid and Offer price (EBBO)

The following message content is used for dissemination of EuroCTP's EBBO message.

MarketDataSnapshotFullRefresh [W] – Message				
Tag	Field	Description	Supported values	Comment/Format
	<StandardHeader> component	MsgType = W		
34	MsgSeqNum	Instrument Level Sequence Number		
	<Instrument> component			
start Instrument				

55	Symbol		N/A	Required when using FIX TagValue encoding (ISO 3531-1) and Instrument component is marked as required. Set to "[N/A]" in such a case.
48	SecurityID	Instrument identification code	Alphanumerical 12	ISIN code ISO 6166
22	SecurityIDSource		4=ISIN	
207	SecurityExchange	Code used to identify the market on which the instrument is listed.	MIC	MIC code ISO 10383
end Instrument				
3102	MostLiquidMarketID	MRMTL Most liquid market in terms of liquidity	MIC	MIC code ISO 10383
	<MDFullGrp> group			Repeating group, one entry for the bid and one entry for the offer
start MDFullGrp				
268	NoMDEntries			
→ 269	MDEntryType	Best bid and Best bid volume Best offer and Best offer volume	0 = Bid 1 = Offer J = Empty book j = Uncrossing bid k = Uncrossing offer	
→ 270	MDEntryPx	Best bid when combined with MDEntryType = 0 Best offer when combined with MDEntryType = 1 Uncrossing bid if MDEntryType = j Uncrossing offer if MDEntryType = k	Decimal	
→ 15	Currency	Currency		Currency code ISO 4217

		Major currency unit in which the price is expressed (applicable if the price is expressed as monetary value).		
→ 271	MDEntrySize	Best bid volume when combined with MDEntryType = 0 Best offer volume when combined with MDEntryType = 1	Decimal	
	<TrdRegTimestamps> group			
→ start <TrdRegTimestamps> group				
→ 768	NoTrdRegTimestamps			
→ → 769	TrdRegTimestamp	Timestamp	Timestamp	Nanoseconds from start of Epoch
→ → 770	TrdRegTimestampType		2 = Time in - Reception date and time when combined with TrdRegTimestampOrigin = P 11 = Publicly reported – Dissemination date and time when combined with TrdRegTimestampOrigin = P 11 = Publicly reported - Publication date and time when combined with TrdRegTimestampOrigin = C 34 = Reference time for BBO - EBBO timestamp when combined with TrdRegTimestampOrigin = P 36 = Update time - Update date and time when combined with TrdRegTimestampOrigin = C	

→ → 771	TrdRegTimestampOrigin		C = Contributor – Trading venue P = Publisher	
→ end <TrdRegTimestamps> group				
→ 276	QuoteConditions		0 = none, E = Locked o = Out of sequence 8 = Crossed due to latest bid 9 = Crossed due to latest offer	
→ 278	MDEntryID	EBBO Identifier		Alphanumerical
end MDFullGrp				
	<StandardTrailer> component			

FIX SBE message structure for European Best Bid and Offer message (RTS on input and output data of consolidated tapes references are in **bold**)

MarketDataSnapshotFullRefresh [W] – Message				
Logical FIX element	SBE data type	#bytes	Format	Comment/**ESMA table reference (Annex III Table 3)**
StandardHeader	N/A	0		Do Not use the FIX Session Layer
MsgSeqNum (34)	SeqNum	8		Instrument Level Sequence Number
Symbol (55)	char	0		Always "[N/A]" (constant value)
SecurityID (48)	char	12	ISIN code ISO 6166	**Instrument identification code Field 2**
SecurityIDSource(22)	uint8	0		Always 4=ISIN (constant value)
SecurityExchange (207)	char	4		MIC code ISO 10383/Instrument place of listing
MostLiquidMarketID (3102)	char	4		MIC code ISO 10383 / **MRMTL Field 7**
NoMDEntries (268)	uint8	0		Always 2 (constant value)
MDEntryType_0 (269)	char	1	0, J or j	0 – Bid J – Empty book j – Uncrossing bid

MDEntryPx_0 (270)	Decimal	9		**Best bid Field 4** when combined with MDEntryType_0 = 0
Currency_0 (15)	char	3		Currency code ISO 4217 / **Price currency Field 3**
MDEntrySize_0 (271)	Decimal	9		**Best bid volume Field 5** when combined with MDEntryType_0 = 0
MDEntryType_1 (269)	char	1	1, J or k	1 - Offer J - Empty book k - Uncrossing offer
MDEntryPx_1 (270)	Decimal	9		**Best offer Field 8** when combined with MDEntryType_1 = 1
Currency_1 (15)	char	3		Currency code ISO 4217 / **Price currency Field 3**
MDEntrySize_1 (271)	Decimal	9		**Best offer volume Field 9** when combined with MDEntryType = 1
QuoteConditions (276)	uint8	1	0, 1, 2, 3, 4, 6, 8 or 10	0 = none, 1 = E - Locked 2 = o - Out of sequence 4 = 8 - Crossed due to latest bid 8 = 9 - Crossed due to latest offer
MDEntryID (278)	InternalIdentifier	8		EBBO identifier
NoTrdRegTimestamps (768)	uint8	0		Always 4 (constant value)
TrdRegTimestamp_0 (769)	Timestamp	8		Nanoseconds from start of Epoch
TrdRegTimestampType_0 (770)	uint8	0		Always 2 - Time in - Reception date and time when combined with TrdRegTimestampOrigin = P (constant value)
TrdRegTimestampOrigin_0 (771)	char	0		Always P - Publisher (constant value)
TrdRegTimestamp_1 (769)	Timestamp	8		Nanoseconds from start of Epoch
TrdRegTimestampType_1 (770)	uint8	0		Always 11 - Publicly reported - **Dissemination date and time Field 10** when combined with TrdRegTimestampOrigin = P (constant value)
TrdRegTimestampOrigin_1 (771)	char	0		Always P – Publisher (constant value)
TrdRegTimestamp_2 (769)	Timestamp	8		Nanoseconds from start of Epoch

TrdRegTimestampType_2 (770)	uint8	O		Always 11 - Publicly reported - **Publication date and time Field 11** when combined with TrdRegTimestampOrigin = C (constant value)
TrdRegTimestampOrigin_2 (771)	char	O		Always C - Contributor - Trading venue (constant value)
TrdRegTimestamp_3 (769)	Timestamp	8		Nanoseconds from start of Epoch
TrdRegTimestampType_3 (770)	uint8	O		Always 34 - Reference time for NBBO - **EBBO timestamp Field 6** when combined with TrdRegTimestampOrigin = P (constant value)
TrdRegTimestampOrigin_3 (771)	char	O		Always P – Publisher (constant value)
TrdRegTimestamp_4 (769)	Timestamp	8		Nanoseconds from start of Epoch
TrdRegTimestampType_4 (770)	uint8	O		Always 36 - Update time - **Entry date and time Field 1** when combined with TrdRegTimestampOrigin = C (constant value)
TrdRegTimestampOrigin_4 (771)	char	O		Always C - Contributor - Trading venue (constant value)
StandardTrailer	N/A	O		Do not use the FIX Session Layer

4.2.9. European Best Auction Price (EBAP)

The following message content is used for dissemination of EuroCTP's EBAP message.

MarketDataSnapshotFullRefresh [W] - Message				
Tag	Field	Description	Supported values	Comment/Format
	<StandardHeader> component	MsgType = W		
34	MsgSeqNum	Instrument Level Sequence Number		
	<Instrument> component			

start Instrument				
55	Symbol		N/A	Required when using FIX TagValue encoding (ISO 3531-1) and Instrument component is marked as required. Set to "[N/A]" in such a case.
48	SecurityID	Instrument identification code	Alphanumerical 12	ISIN code ISO 6166
22	SecurityIDSource		4=ISIN	
207	SecurityExchange	Code used to identify the market on which the instrument is listed.	MIC	MIC code ISO 10383
end Instrument				
3102	MostLiquidMarketID	MRMTL Most liquid market in terms of liquidity	MIC	MIC code ISO 10383
	<MDFullGrp> group			Repeating group, one entry for the bid and one entry for the offer
start MDFullGrp				
268	NoMDEntries			
→ 269	MDEntryType	Auction clearing information to provide auction information Empty book to remove auction price information	Q = Auction clearing price J = Empty book	
→ 270	MDEntryPx	Volume weighted auction price	Decimal	
→ 15	Currency	Currency Major currency unit in which the price is expressed (applicable if the price is expressed as monetary value).		Currency code ISO 4217
→ 271	MDEntrySize	Auction volume	Decimal	
→ 333	LowPx	Lowest auction price	Decimal	

→ 332	HighPx	Highest auction price	Decimal	
	<TrdRegTimestamps> group			
→ start <TrdRegTimestamps> group				
→ 768	NoTrdRegTimestamps			
→ → 769	TrdRegTimestamp	Timestamp	Timestamp	Nanoseconds from start of Epoch
→ → 770	TrdRegTimestampType	2, 11, 34 or 36	2 = Time in - Reception date and time when combined with TrdRegTimestampOrigin = P 11 = Publicly reported - Dissemination date and time when combined with TrdRegTimestampOrigin = P 11 = Publicly reported - Publication date and time when combined with TrdRegTimestampOrigin = C 34 - Reference time for NBBO - EBAP calculation timestamp when combined with TrdRegTimestampOrigin = P 36 = Update time - Indicative date and time when combined with TrdRegTimestampOrigin = C	
→ → 771	TrdRegTimestampOrigin		C = Contributor - Trading venue P = Publisher	
→ end <TrdRegTimestamps> group				
QuoteConditions (276)	QuoteConditions		0 = none, o = Out of sequence	

→ 278	MDEntryID	EBAP Identifier		Alphanumerical
end MDFullGrp				
	<StandardTrailer> component			

FIX SBE message structure for European Best Auction Price message (RTS on input and output data of consolidated tapes references are in **bold**).

MarketDataSnapshotFullRefresh [W] – Message				
Logical FIX element	SBE data type	#bytes	Format	Comment/**ESMA table reference (Annex III Table 4)**
StandardHeader	N/A	0		Do Not use the FIX Session Layer
MsgSeqNum (34)	SeqNum	8		Instrument Level Sequence Number
Symbol (55)	char	0		Always "[N/A]" (constant value)
SecurityID (48)	char	12	ISIN code ISO 6166	**Instrument identification code Field 2**
SecurityIDSource(22)	uint8	0		Always 4 – ISIN (constant value)
SecurityExchange (207)	char	4	MIC code ISO 10383	MIC code ISO 10383/Instrument place of listing
MostLiquidMarketID (3102)	char	4	MIC code ISO 10383	**MRMTL Field 8**
NoMDEntries (268)	uint8	0		Always 1 – no repeating groups (constant value)
MDEntryType (269)	char	1	Q or J	Q – Auction clearing price J – Empty book
MDEntryPx (270)	Decimal	9		**Volume weighted Auction Price Field 3c** when combined with MDEntryType = Q
Currency (15)	char	3		Currency code ISO 4217 / **Price currency Field 4**
MDEntrySize (271)	Decimal	9		**Auction volume Field 5** when combined with MDEntryType = Q
LowPx (333)	Decimal	9		**Lowest auction price Field 3a**
HighPx (332)	Decimal	9		**Highest auction price Field 3b**

QuoteConditions (276)	uint8	1	0 or 2	0 = none, 2 = o - Out of sequence
MDEntryID (278)	InternalIdentifier	8		EBAP identifier
NoTrdRegTimestamps (768)	uint8	0		Always 5 (constant value)
TrdRegTimestamp_0 (769)	Timestamp	8		Nanoseconds from start of Epoch
TrdRegTimestampType_0 (770)	uint8	0		Always 2 - Time in - Reception date and time when combined with TrdRegTimestampOrigin = P (constant value)
TrdRegTimestampOrigin_0 (771)	char	0		Always P – Publisher (constant value)
TrdRegTimestamp_1 (769)	Timestamp	8		Nanoseconds from start of Epoch
TrdRegTimestampType_1 (770)	uint8	0		Always 11 - Publicly reported - Dissemination date and time when combined with TrdRegTimestampOrigin = P (constant value)
TrdRegTimestampOrigin_1 (771)	char	0		Always P – Publisher (constant value)
TrdRegTimestamp_2 (769)	Timestamp	8		Nanoseconds from start of Epoch
TrdRegTimestampType_2 (770)	uint8	0		Always 11 - Publicly reported - Publication date and time when combined with TrdRegTimestampOrigin = C (constant value)
TrdRegTimestampOrigin_2 (771)	char	0		Always C - Contributor - Trading venue (constant value)
TrdRegTimestamp_3 (769)	Timestamp	8		Nanoseconds from start of Epoch
TrdRegTimestampType_3 (770)	uint8	0		Always 34 - Reference time for NBBO - EBAP calculation timestamp when combined with TrdRegTimestampOrigin = P (constant value)
TrdRegTimestampOrigin_3 (771)	char	0		Always P – Publisher (constant value)
TrdRegTimestamp_4 (769)	Timestamp	8		Nanoseconds from start of Epoch
TrdRegTimestampType_4 (770)	uint8	0		Always 36 - Update time - Indicative date and time when combined with TrdRegTimestampOrigin = C (constant value)
TrdRegTimestampOrigin_4 (771)	char	0		Always C - Contributor - Trading venue (constant value)

StandardTrailer	N/A	O		Do not use the FIX Session Layer

4.2.10.European Frequent Batch Auction (EFBA)

The following message content is used for dissemination of EuroCTP's EFBA message.

MarketDataSnapshotFullRefresh [W] – Message				
Tag	Field	Description	Supported values	Comment/Format
	<StandardHeader> component	MsgType = W		
34	MsgSeqNum	Instrument Level Sequence Number		
	<Instrument> component			
start Instrument				
55	Symbol		N/A	Required when using FIX TagValue encoding (ISO 3531-1) and Instrument component is marked as required. Set to "[N/A]" in such a case.
48	SecurityID	Instrument identification code	Alphanumerical 12	ISIN code ISO 6166
22	SecurityIDSource		4=ISIN	
207	SecurityExchange	Code used to identify the market on which the instrument is listed.	MIC	MIC code ISO 10383
end Instrument				
3102	MostLiquidMarketID	MRMTL Most liquid market in terms of liquidity	MIC	MIC code ISO 10383
	<MDFullGrp> group			Repeating group, one entry for the bid and one entry for the offer
start MDFullGrp				

268	NoMDEntries			
→ 269	MDEntryType	Auction clearing information to provide auction information Empty book to remove auction price information	Q = Auction clearing price J = Empty book	
→ 270	MDEntryPx	Volume weighted auction price	Decimal	
→ 15	Currency	Currency Major currency unit in which the price is expressed (applicable if the price is expressed as monetary value).		Currency code ISO 4217
→ 271	MDEntrySize	Auction volume	Decimal	
→ 333	LowPx	Lowest auction price	Decimal	
→ 332	HighPx	Highest auction price	Decimal	
→ 336	TradingSessionID		1= Day (constant value)	
→ 625	TradingSessionSubID	Trading System Phase To indicate Frequent Batch Auction	Order initiated auction - Use for Trading System Phase 'ODAU'	
	<TrdRegTimestamps> group			
→ start <TrdRegTimestamps> group				
→ 768	NoTrdRegTimestamps			
→ → 769	TrdRegTimestamp	Timestamp	Timestamp	Nanoseconds from start of Epoch
→ → 770	TrdRegTimestampType	2, 11, 34 or 36	2 = Time in - Reception date and time when combined	

			with TrdRegTimestampOrigin = P 11 = Publicly reported – Dissemination date and time when combined with TrdRegTimestampOrigin = P 11 = Publicly reported – Publication date and time when combined with TrdRegTimestampOrigin = C 34 = Reference time for NBBO – EFBA calculation timestamp when combined with TrdRegTimestampOrigin = P 36 = Update time – Indicative date and time when combined with TrdRegTimestampOrigin = C	
→ → 771	TrdRegTimestampOrigin		C = Contributor – Trading venue P = Publisher	
→ end <TrdRegTimestamps> group				
→ 276	QuoteConditions		0 = none o = 'Out of sequence'	
→ 278	MDEntryID	EFBA Identifier		Alphanumerical
end MDFullGrp				
	<StandardTrailer> component			

FIX SBE message structure for European Frequent Batch Auction message (RTS on input and output data of consolidated tapes references are in **bold**)

MarketDataSnapshotFullRefresh [W] – Message				
Logical FIX element	SBE data type	#bytes	Format	Comment/**ESMA table reference (Annex III Table 4)**
StandardHeader	N/A	0		Do Not use the FIX Session Layer
MsgSeqNum (34)	SeqNum	8		Instrument Level Sequence Number
Symbol (55)	char	0		Always "[N/A]" (constant value)
SecurityID (48)	char	12	ISIN code ISO 6166	**Instrument identification code Field 2**
SecurityIDSource(22)	uint8	0		Always 4=ISIN (constant value)
SecurityExchange (207)	char	4	MIC code ISO 10383	MIC code ISO 10383/Place of listing
MostLiquidMarketID (3102)	char	4	MIC code ISO 10383	**MRMTL Field 8**
NoMDEntries (268)	uint8	0		Always 1 – no repeating groups (constant value)
MDEntryType (269)	char	1	Q or J	Q – Auction clearing price J – Empty book
MDEntryPx (270)	Decimal	9		**Volume weighted Auction Price Field 3c** when combined with MDEntryType = Q
Currency (15)	char	3		Currency code ISO 4217 / **Price currency Field 4**
MDEntrySize (271)	Decimal	9		**Auction volume Field 5** when combined with MDEntryType = Q
TradingSessionID (336)	uint8	0		Always 1= Day (constant value)
TradingSessionSubID (625)	uint8	0		Always 14 – Order initiated auction Trading phase 'ODAU' (constant value)
LowPx (333)	Decimal	9		**Lowest auction price Field 3a**
HighPx (332)	Decimal	9		**Highest auction price Field 3b**
QuoteConditions (276)	uint8	1	0 or 2	0 = none 2 = o – 'Out of sequence'

MDEntryID (278)	InternalIdentifier	8		EFBA identifier
NoTrdRegTimestamps (768)	uint8	0		Always 5 (constant value)
TrdRegTimestamp_0 (769)	Timestamp	8		Nanoseconds from start of Epoch
TrdRegTimestampType_0 (770)	uint8	0		Always 2 – Time in – Reception date and time when combined with TrdRegTimestampOrigin = P (constant value)
TrdRegTimestampOrigin_0 (771)	char	0		Always P – Publisher (constant value)
TrdRegTimestamp_1 (769)	Timestamp	8		Nanoseconds from start of Epoch
TrdRegTimestampType_1 (770)	uint8	0		Always 11 – Publicly reported – **Dissemination date and time Field 6** when combined with TrdRegTimestampOrigin = P (constant value)
TrdRegTimestampOrigin_1 (771)	char	0		Always P – Publisher (constant value)
TrdRegTimestamp_2 (769)	Timestamp	8		Nanoseconds from start of Epoch
TrdRegTimestampType_2 (770)	uint8	0		Always 11 – Publicly reported – **Publication date and time Field 7** when combined with TrdRegTimestampOrigin = C (constant value)
TrdRegTimestampOrigin_2 (771)	char	0		Always C – Contributor – Trading venue (constant value)
TrdRegTimestamp_3 (769)	Timestamp	8		Nanoseconds from start of Epoch
TrdRegTimestampType_3 (770)	uint8	0		Always 34 – Reference time for NBBO – EFBA calculation timestamp when combined with TrdRegTimestampOrigin = P (constant value)
TrdRegTimestampOrigin_3 (771)	char	0		Always P – Publisher (constant value)
TrdRegTimestamp_4 (769)	Timestamp	8		Nanoseconds from start of Epoch

TrdRegTimestampType_4 (770)	uint8	O		Always 36 - Update time - **Indicative date and time Field 1** when combined with TrdRegTimestampOrigin = C (constant value)
TrdRegTimestampOrigin_4 (771)	char	O		Always C - Contributor - Trading venue (constant value)
StandardTrailer	N/A	O		Do not use the FIX Session Layer

4.2.11. Post-trade

The following message structure is used for dissemination of EuroCTP's post-trade message. MarketDataIncrementalRefresh (35=X)

MarketDataIncrementalRefresh [X] - Message				
Tag	Field	Description	Supported values	Comment/Format
	<StandardHeader> component	MsgType = X		
	<MDIncGrp> group			
start MDIncGrp				
268	NoMDEntries			
→ 279	MDUpdateAction	Flags Used when amending or cancelling a previously reported transaction.	0 = New 1 = Change 'AMND' 2 = Delete 'CANC'	
→ 269	MDEntryType		2 = Trade	
→ 34	MsgSeqNum	Instrument Level Sequence Number		
	<Instrument component>			
→ *start <Instrument> component*				
→ 55	Symbol		N/A	Required when using FIX TagValue encoding (ISO

				3531-1) and Instrument component is marked as required. Set to "[N/A]" in such a case.
→ 48	SecurityID	Instrument identification code	Alphanumerical 12	ISIN code ISO 6166
→ 22	SecurityIDSource		4=ISIN	
→ 207	SecurityExchange	Code used to identify the market on which the instrument is listed.	MIC	MIC code ISO 10383
→ end <Instrument> component				
→ 30	LastMkt	Venue of execution Identification of the venue where the transaction was executed (segment MIC or operating MIC where segment MIC does not exist)	MIC	MIC code ISO 10383
→ 270	MDEntryPx	Traded price of the transaction excluding, where applicable, commission and accrued interest. Where price is currently not available but pending ('PNDG') or not applicable ('NOAP'), this field shall not be populated.		
→ 15	Currency	Price currency Major currency unit in which the price is expressed (applicable if the price is expressed as monetary value).		Currency code ISO 4217
→ 271	MDEntrySize	Quantity Number of units of the financial instruments.		
→ 1024	MDOriginType	Trading system Type of trading system on which the instrument is traded.	Omit when LastMkt(30) is set to 'SINT' or 'XOFF'. 0 = Central limit order book – Trading System Type 'CLOB'.	

			3 = Quote driven market – Trading System Type 'QDTS'. 5 = Auction driven market – Trading System Type 'PATS'. 6 = Quote negotiation – Trading System Type 'RFQT' 8 = Hybrid market – Trading System type 'HYBR' 9 = Other market – Trading System Type 'OTHR'	
→ 2667	AlgorithmicTradeIndicator		0 = Non-algorithmic trade 1 = Algorithmic trade 'ALGO'	
→ 3105	MDQualityIndicator	Suspicious trade flag bitmap mask Data quality flag reported by the CTP when the APA or the CTP have identified trades that for them might be subject to data quality issues.	0 = No data quality issue 'FALSE' 1 = Suspicious trade 'TRUE' 2 = Suspicious trade due to size deviation 'TRUE' 4 = Suspicious trade due to price deviation 'TRUE'	
	<RegulatoryTradeIDGrp> group			
→ start <RegulatoryTradeIDGrp> group				
→ 1907	NoRegulatoryTradeIDs			
→ → 1903	RegulatoryTradeID	Transaction identification code Alphanumerical code assigned by trading venues (pursuant to Article 12 of Commission Delegated Regulation (EU) 2017/580 and APAs and used in any subsequent reference to the specific trade.**		
→ → 1906	RegulatoryTradeIDType	Type of transaction identifier	5 = Trading venue transaction identifier	
end <RegulatoryTradeIDGrp> group				

	<TrdRegTimestamps> group			
→ *start <TrdRegTimestamps> group*				
→ 768	NoTrdRegTimestamps			
→ → 769	TrdRegTimestamp	Timestamp	Timestamp	Nanoseconds from start of Epoch
→ → 770	TrdRegTimestampType		1 = Execution time – Trading date and time when combined with • TrdRegTimestamp Origin = C 2 = Time in – Reception date and time • Reception by the CTP when combined with TrdRegTimestamp Origin = P • Reception by the APA when combined with TrdRegTimestamp Origin = C 11 = Publicly reported – Publication date and time • Publication by the CTP when combined with TrdRegTimestamp Origin = P • Publication by the Data Contributor when combined with TrdRegTimestamp Origin = C	
→ → 771	TrdRegTimestampOrigin		C = Contributor – Trading venue P = Publisher	
→ *end <TrdRegTimestamps> group*				
	<Parties> group			
→ *start <Parties> group*				

→ 453	NoPartyIDs			
→ → 448	PartyID	Venue of publication MIC code when combined with PartyIDSource = 62 Third-country trading venue of execution (when LastMkt(30) = 'XOFF') MIC code when combined with PartyIDSource = 73		MIC code ISO 10383
→ → 447	PartyIDSource	MIC code	G = Market Identifier Code	
→ → 452	PartyRole		62 = Report originator - Venue of publication 73 = Execution venue - Third-country trading venue of execution	
→ end <Parties> group				
	<TradeTypeGrp> group			
→ start <TradeTypeGrp> group				
→ 3005	NoTradeTypes			
→ → 3006	TradeType	Flags	50 = Portfolio trade - 'PORT' 64 = Benchmark - 'BENC' 65 = Package trade - 'CONT'	
→ end <TradeTypeGrp> group				
	<TradePriceConditionGrp> group			
→ start <TradePriceConditionGrp> group				
→ 1838	NoTradePriceConditions			
→ → 1839	TradePriceCondition	Flags	13 = Special dividend - 'SDIV' 15 = Non-price forming trade - 'NPFT' 17 = Price or strike price is pending - 'PNDG' 18 = Price is not applicable - 'NOAP'	

→ *end <TradePriceConditionGrp> group*				
	<TrdRegPublicationGrp> group			
→ *start <TrdRegPublicationGrp> group*				
→ 2668	NoTrdRegPublications			
→ → 2669	TrdRegPublicationType		0 = Pre-trade transparency waiver • 'NLIQ' when combined with TrdRegPublication Reason = 0 • 'OILQ' when combined with TrdRegPublication Reason = 1 • 'PRIC' when combined with TrdRegPublication Reason = 2 • 'RFPT' when combined with TrdRegPublication Reason = 3 1 = Post-trade deferral - 'LRGS'	
→ → 2670	TrdRegPublicationReason		0 = No preceding order in book as transaction price sit within average spread of a liquid instrument - 'NLIQ' 1= No preceding order in book as transaction price depends on system-set reference price for an illiquid instrument - 'OILQ' 2 = No preceding order in book as transaction price if for transaction subject to conditions other than current market price - 'PRIC' 3 = No public price for preceding order as public reference price was used for matching orders - 'RFPT'	

			6 = Deferral due to “Large in Scale” – ‘LRGS’	
→ end <TrdRegPublicationGrp> group				
→ 8031	MMTCode	MMT efficient encoding flag		14 character string
→ 277	TradeCondition	Out of sequence flag	0 =‘FALSE’, k = ‘TRUE’	
end MDIncGrp				
	<StandardTrailer> component			

FIX SBE message structure for Post-trade message (RTS on input and output data of consolidated tapes references are in **bold**)

MarketDataIncrementalRefresh [X] – Message				
Logical FIX element	SBE data type	#bytes	Format	Comment/**ESMA table reference (Annex II Table 7)**
StandardHeader	N/A	0		Do Not use the FIX Session Layer
NoMDEntries (268)	uint8	0		Always 1 – no repeating groups (constant value)
MDUpdateAction(279)	uint8	1		0 = New 1 = Change ‘AMND’ **Flags Field 16** 2 = Delete ‘CANC’ **Flags Field 16**
MDEntryType(269)	uint8	0		Always 2 = Trade (constant value)
MsgSeqNum (34)	SeqNum	8		Instrument Level Sequence Number
Symbol (55)	char	0		Always N/A (constant value)
SecurityID (48)	char	12	ISIN code ISO 6166	Instrument or Composite ISIN code / **Instrument identification code Field 2**
SecurityIDSource (22)	uint8	0		Always 4 – ISIN (constant value)
SecurityExchange (207)	char	4	MIC code ISO 10383	MIC code ISO 10383/Place of listing
LastMkt (30)	char	4	MIC code ISO 10383	MIC code / **Venue of execution Field 7**
MDEntryPx (270)	Decimal	9		Floating point decimal / **Price Field 3**

Currency (15)	char	3	CUR code ISO 4217	**Price currency Field 5**
MDEntrySize(271)	Decimal	9		Floating point decimal / **Quantity Field 6**
MDOriginType(1024)	uint8	1	0, 3, 5, 6, 8 or 9	**Trading system Field 10** Omit when LastMkt(30) is set to 'SINT' or 'XOFF'. 0 = Central limit order book – Trading System Type 'CLOB'. 3 = Quote driven market – Trading System Type 'QDTS'. 5 = Auction driven market – Trading System Type 'PATS'. 6 = Quote negotiation – Trading System Type 'RFQT' 8 = Hybrid market – Trading System type 'HYBR' 9 = Other market – Trading System Type 'OTHR'
AlgorithmicTradeIndicator(2667)	uint8	1	0 or 1	0 = Non-algorithmic trade 1 = Algorithmic trade 'ALGO' **Flags Field 16**
MDQualityIndicator(3105)	uint8	1	0, 1, 2, or 4	**Suspicious data flag Field 17** 0 = No data quality issue 'FALSE' 1 = Suspicious trade 'TRUE' 2 = Suspicious trade due to size deviation 'TRUE' 4 = Suspicious trade due to price deviation 'TRUE'
NoRegulatoryTradeIDs (1907)	uint8	0		Always 1 (constant value)
RegulatoryTradeID (1903)	char	52		(up to 52 alphanumerical characters) **Transaction identification code Field 13**
RegulatoryTradeIDType (1906)	uint8	0		Always 5 – TVTIC (constant value)
NoTradeTypes (3005)	uint8	0		Use bitmap for repeating group content
TradeType (3006)	uint8	1	Combination of bits	Bitmap with the following choices 1 = 50 – Portfolio trade – 'PORT' **Flags Field 16** 2 = 64 – Benchmark – 'BENC' **Flags Field 16** 4 = 65 – Package trade – 'CONT' **Flags Field 16**
NoTradePriceConditions (1838)	uint8	0		Use bitmap for repeating group content
TradePriceCondition (1839)	uint8	1	Combination of bits	Bitmap with the following choices 1 = 13 – Special dividend 'SDIV' **Flags Field 16** 2 = 15 – Non-price forming trade 'NPFT' **Flags Field 16** 4 = 17 – Price or strike price is pending 'PNDG' **Missing price Field 4**

				8 = 18 - Price not applicable 'NOAP' **Missing price Field 4**
NoTrdRegPublications (2668)	uint8	0		Use bitmap for repeating group content
TrdRegPublicationType (2669)	uint8	0		Always 0 - Pre-trade transparency waiver except for 'LRGS' **Flags Field 16** where it is 1 - Post-trade deferral
TrdRegPublicationReason (2670)	uint8	1	Combination of bits	Bitmap with the following choices 1 = 'NLIQ' **Flags Field 16** 2 = 'OILQ' **Flags Field 16** 4 = 'PRIC' **Flags Field 16** 8 = 'RFPT' **Flags Field 16** 16 = 'LRGS' **Flags Field 16**
MMTCode (8031)	char	14		MMTCode
TradeCondition (277)	char	1	0 or k	0 = none k - 'TRUE'
NoTrdRegTimestamps (768)	uint8	0		Always 5 (constant value)
TrdRegTimestamp_0 (769)	Timestamp	8		Nanoseconds from start of Epoch
TrdRegTimestampType_0 (770)	uint8	0		Always 1 - Execution time - **Trading date and time Field 1** when combined with TrdRegTimestampOrigin = C (constant value)
TrdRegTimestampOrigin_0 (771)	char	0		Always C - Contributor (constant value)
TrdRegTimestamp_1 (769)	Timestamp	8		Nanoseconds from start of Epoch
TrdRegTimestampType_1 (770)	uint8	0		Always 2 -Time in - Reception by the CTP **Field 14** when combined with TrdRegTimestampOrigin = P (constant value)
TrdRegTimestampOrigin_1 (771)	char	0		Always P - Publisher (constant value)
TrdRegTimestamp_2 (769)	Timestamp	8		Nanoseconds from start of Epoch
TrdRegTimestampType_2 (770)	uint8	0		Always 2 - Time in - Reception by the APA **Field 9** when combined with TrdRegTimestampOrigin = C (constant value)
TrdRegTimestampOrigin_2 (771)	char	0		Always C - Contributor (constant value)
TrdRegTimestamp_3 (769)	Timestamp	8		Nanoseconds from start of Epoch

TrdRegTimestampType_3 (770)	uint8	0		Always 11 - Publication by the CTP **Field 15** when combined with TrdRegTimestampOrigin = P (constant value)
TrdRegTimestampOrigin_3 (771)	char	0		Always P - Publisher (constant value)
TrdRegTimestamp_4 (769)	Timestamp	8		Nanoseconds from start of Epoch
TrdRegTimestampType_4 (770)	uint8	0		Always 11 - Publicly reported - Publication by the Data Contributor **Field 11** when combined with TrdRegTimestampOrigin = C (constant value)
TrdRegTimestampOrigin_4 (771)	char	0		Always C = Contributor (constant value)
NoPartyIDs (453)	RptGrpHeader	2		Standard SBE group header with entry length and entry size
PartyID (448)	char	4	MIC code ISO 10383	**Venue of publication Field 12** when combined with PartyRole = 62 **Third-country trading venue of execution Field 8** when combined with PartyRole = 73
PartyIDSource (447)	char	0		Always G - MIC (constant value)
PartyRole (452)	uint8	1	62 or 73	62 - Report originator 73 - Execution venue
StandardTrailer	N/A	0		Do not use the FIX Session Layer

4.2.12. Instrument status

The following message structure is used for dissemination of EuroCTP's Instrument status. SecurityStatus (35=f)

SecurityStatus [f] - Message				
Tag	Field	Description	Supported values	Comment/Format
	<StandardHeader> component	MsgType = f		Includes the sequence number
34	MsgSeqNum	Instrument Level Sequence Number		

	<Instrument> component			ISIN code ISO 6166
start Instrument				
55	Symbol		N/A	Required when using FIX TagValue encoding (ISO 3531-1) and Instrument component is marked as required. Set to "[N/A]" in such a case.
48	SecurityID	Instrument identification code	Alphanumerical 12	ISIN code ISO 6166
22	SecuirtyIDSource		4=ISIN	
207	SecurityExchange	Code used to identify the market on which the instrument is listed.	MIC	MIC code ISO 10383
965	SecurityStatus	Instrument status (except 'HALT')	0 = empty 1 = Active - Instrument status 'ACTV' when there is no previous status. 5 = Delisted status 'RMOV' 9 = Suspended status 'SUSP'	
end instrument				
15	Currency	Major currency in which the instrument trades.	Currency	Currency code ISO 4217
1301	MarketID	Trading venue	MIC	MIC code ISO 10383
3103	MostLiquidMarketIndicator	Most Relevant Market in terms of liquidity	0-'FALSE', 1-'TRUE'	
1430	VenueType	Trading system Type of trading system on which the instrument is traded.	B = Central limit order book - Trading System Type 'CLOB'. Q = Quote driven market - Trading System Type 'QDTS'. A = Auction driven market - Trading System Type 'PATS'. N = Quote negotiation - Trading System Type 'RFQT'	

			H = Hybrid market – Trading System type 'HYBR' z = Other market – Trading System Type 'OTHR'	
336	TradingSessionID		1= Day	
625	TradingSessionSubID	Trading system phase Mutually exclusive with MatchType (574)	0 = empty 8 = Undefined Auction 'UDUC' 2 = Scheduled Opening Auction 'SOAU' 4 = Scheduled Closing Auction 'SCAU' 6 = Scheduled Intraday Auction 'SIAU' 9 = Unscheduled Auction 'UAUC' 14 = On Demand Auction (Frequent Batched Auction) 'ODAU' 3 = Continuous Trading 'COTR' 5 = At Market Close Trading 'MACT' 10 = Out of Main Session Trading 'OMST' 99 = Other 'OTSP'	
574	MatchType	Trading system phase Mutually exclusive with TradingSessionSubID(625).	0 = empty 1 = Trade Reporting (Off Exchange) 'TROF' 3 = Trade Reporting (On Exchange) 'TROE' 9 = Trade Reporting (Systematic Internaliser) 'TRSI'	
326	SecurityTradingStatus	Instrument status Use 3=Resume when state changes to 'ACTV' after being in 'SUSP' or 'HALT'.	0 = empty 2 = Trading halt 'HALT'. 3 = Resume – Instrument Status 'ACTV' when previous status was 'SUSP' or 'HALT'.	
60	TransactTime	Dissemination date and time Date and time on which the regulatory data is disseminated by the CTP		Nanoseconds from start of Epoch

168	EffectiveTime	Instrument status start date and time Date and time from which the instrument status is valid.		Nanoseconds from start of Epoch
278	MDEntryID	Instrument status identifier		
	<StandardTrailer> component			

FIX SBE message structure for Instrument status (RTS on input and output data of consolidated tapes references are in **bold**).

SecurityStatus [f] – Message				
Logical FIX element	SBE data type	#bytes	Format	Comment/**ESMA table reference (Annex II Table 4)**
StandardHeader	N/A	0		Do Not use the FIX Session Layer
MsgSeqNum (34)	SeqNum	8		Instrument Level Sequence Number
Symbol(55)	char	0		Always "[N/A]" (constant value)
SecurityID (48)	char	12		**Instrument identification code Field 1** ISIN code ISO 6166
SecuirtyIDSource (22)	uint8	0		Always 4 – ISIN (constant value)
SecurityStatus (965)	uint8	1	0, 1, 5 or 9	**Instrument status Field 5** 0 – empty 1 – Active – 'ACTV' when there is no previous status. 5 – Delisted status 'RMOV' 9 – Suspended status 'SUSP'
Currency (15)	char	3		Currency code ISO 4217 **Currency Field 3**
MarketID (1301)	char	4		MIC code ISO 10383 **Trading venue Field 6**
SecurityExchange (207)	char	4		MIC code ISO 10383/Place of listing
MostLiquidMarketIndicator (3103)	uint8	1	0 or 1	**Most Relevant Market in terms of liquidity Field 9** 0 = 'FALSE' 1 = 'TRUE'
VenueType (1430)	char	1	B, Q, A, N, H. z	**Trading system Field 7** B – Central limit order book – Trading System Type 'CLOB'. Q – Quote driven market – Trading System Type 'QDTS'.

				A - Auction driven market - Trading System Type 'PATS'. N - Quote negotiation - Trading System Type 'RFQT' H - Hybrid market - Trading System type 'HYBR' z - Other market - Trading System Type 'OTHR'
TradingSessionID (336)	uint8	0		Always 1 - Day (constant value)
TradingSessionSubID (625)	uint8	1	0, 8, 2, 4, 6, 9, 14, 3, 5, 10, 99	**Trading system phase Field 8** 0 – empty 8 - Undefined Auction 'UDUC' 2 - Scheduled Opening Auction 'SOAU' 4 - Scheduled Closing Auction 'SCAU' 6 - Scheduled Intraday Auction 'SIAU' 9 - Unscheduled Auction 'UAUC' 14 - On Demand Auction (Frequent Batched Auction) 'ODAU' 3 - Continuous Trading 'COTR' 5 - At Market Close Trading 'MACT' 10 - Out of Main Session Trading 'OMST' 99 - Other 'OTSP'
MatchType (574)	uint8	1	0, 1, 3 or 9	**Trading system phase Field 8** 0 – empty 1 - Trade Reporting (Off Exchange) 'TROF' 3 - Trade Reporting (On Exchange) 'TROE' 9 - Trade Reporting (Systematic Internaliser) 'TRSI'
SecurityTradingStatus (326)	uint8	1	0, 2 or 3	**Instrument status Field 5** 0 = empty 2 = Trading halt 'HALT'. 3 = Resume - 'ACTV' when previous status was 'SUSP' or 'HALT'.
TransactTime (60)	Timestamp	8		**System status dissemination date and time Field 4** Nanoseconds from start of Epoch
EffectiveTime (168)	Timestamp	8		**Instrument status start date and time Field 2** Nanoseconds from start of Epoch
MDEntryID (278)	InternalIdentifier	8		Instrument status identifier
StandardTrailer	N/A	0		Do not use the FIX Session Layer

4.2.13. Order matching system status

The following message structure is used for dissemination of EuroCTP's Order matching system status. TradingSessionStatus (35=h)

TradingSessionStatus [h] – Message				
Tag	Field	Description	Supported values	Comment/Format
	<StandardHeader> component	MsgType = h		
34	MsgSeqNum	Instrument Level Sequence Number		
1301	MarketID	Trading Venue Identification of the trading venue on which the trading system status is valid (segment MIC where available, otherwise operating MIC).	MIC	MIC code ISO 10383
1430	VenueType	Type of trading system on which the system status is provided.	B = Central limit order book – Use for Trading System Type 'CLOB'. Q = Quote driven market – Use for Trading System Type 'QDTS'. A = Auction driven market – Use for Trading System Type 'PATS'. N = Quote negotiation – Use for Trading System Type 'RFQT' H = Hybrid market – Use for Trading System type 'HYBR' z = Other market – Use for Trading System Type 'OTHR'	
340	TradSesStatus	Status of the trading system	2 = Open – 'ACTV' 7 = Outage – 'OTAG' 8 = Partial outage – 'POTG'	
60	TransactTime	System status dissemination date and time Date and time on which the system status is disseminated by the CTP.		Nanoseconds from start of Epoch

168	EffectiveTime	System status start date and time Date and time from which the system status is valid.		Nanoseconds from start of Epoch
278	MDEntryID	Instrument status identifier		
	<StandardTrailer> component			

FIX SBE message structure for Order Matching System Status message(RTS on input and output data of consolidated tapes references are in **bold**).

TradingSessionStatus [h] - Message				
Logical FIX element	SBE data type	#bytes	Format	Comment/**ESMA table reference (Annex II Table 4)**
StandardHeader	N/A	0		Do Not use the FIX Session Layer
MsgSeqNum (34)	SeqNum	8		Instrument Level Sequence Number
MarketID (1301)	uint8	4		MIC code ISO 10383 / **Trading venue Field 1**
VenueType (1430)	char	1	B, Q, A, N, H, z	**Trading system type Field 2** B – Central limit order book – Use for Trading System Type 'CLOB'. Q – Quote driven market – Use for Trading System Type 'QDTS'. A – Auction driven market – Use for Trading System Type 'PATS'. N – Quote negotiation – Use for Trading System Type 'RFQT' H – Hybrid market – Use for Trading System type 'HYBR' z – Other market – Use for Trading System Type 'OTHR'
TradSesStatus (340)	uint8	1	2, 7 or 8	**Trading system status Field 5** 2 = Open – 'ACTV' 7 = Outage – 'OTAG' 8 = Partial outage – 'POTG'
TransactTime (60)	Timestamp	8		**System status dissemination date and time Field 4** Nanoseconds from start of Epoch
EffectiveTime (168)	Timestamp	8		**System status start date and time Field 3** Nanoseconds from start of Epoch
MDEntryID (278)	InternalIdentifier	8		Instrument status identifier
StandardTrailer	N/A	0		Do not use the FIX Session Layer

5. Contacts

Should you have any questions, please don't hesitate to contact us on product@euroctp.eu

DISCLAIMER

This publication is for information purposes only. The information contained in the document (the "Document") is strictly confidential and has been prepared for information purposes only and does not constitute investment, financial, legal, accounting, tax or other professional advice, nor an offer to sell or a solicitation of an offer to buy any securities, financial instruments or services.
No part of this document may be reproduced, stored in a retrieval system and/or published in any form or by any means, electronic, mechanical, photocopying, recording or by any other means without the prior written permission of EuroCTP.

While we strive to ensure that the information provided is accurate and up-to-date, we make no representations or warranties of any kind, express or implied, about the completeness, accuracy, reliability, suitability, or availability with respect to the presentation or the information, products, services, or related graphics contained in the presentation for any purpose. Nothing contained herein shall be relied upon as a promise or representation whether as to the past or future performance. Any reliance you place on such information is therefore strictly at your own risk. The Document may contain forward-looking statements that involve risks and uncertainties; actual results may differ materially. EuroCTP has no obligation to update, supplement or correct the Document or to notify any person of changes.

EuroCTP expressly disclaims, to the maximum extent permitted by law, any and all liability which may be based on any such information, errors therein or omissions there from and any recipient agrees not to rely thereon.

All product and company names mentioned herein are trademarks or registered trademarks of their respective holders. Use of them does not imply any affiliation with or endorsement by them.
The views and opinions expressed in this document are those of the authors and do not necessarily reflect the official policy or position of EuroCTP.

This disclaimer and any non-contractual obligations arising out of or in connection with it are governed by Dutch law. The competent courts of Amsterdam, the Netherlands, shall have exclusive jurisdiction, without prejudice to rights that cannot be limited under applicable law.